



ECE 252 Project – Phase Three

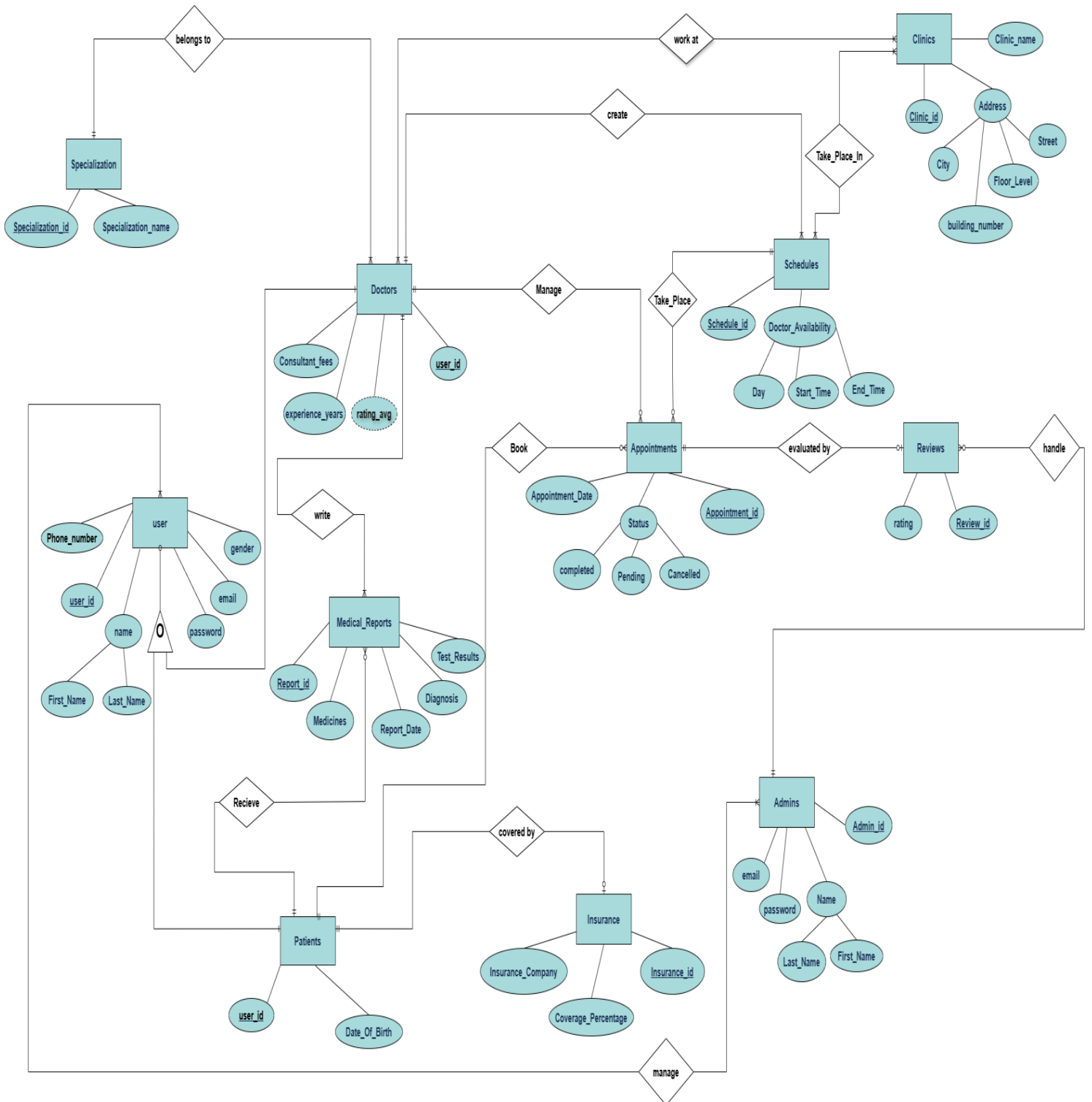
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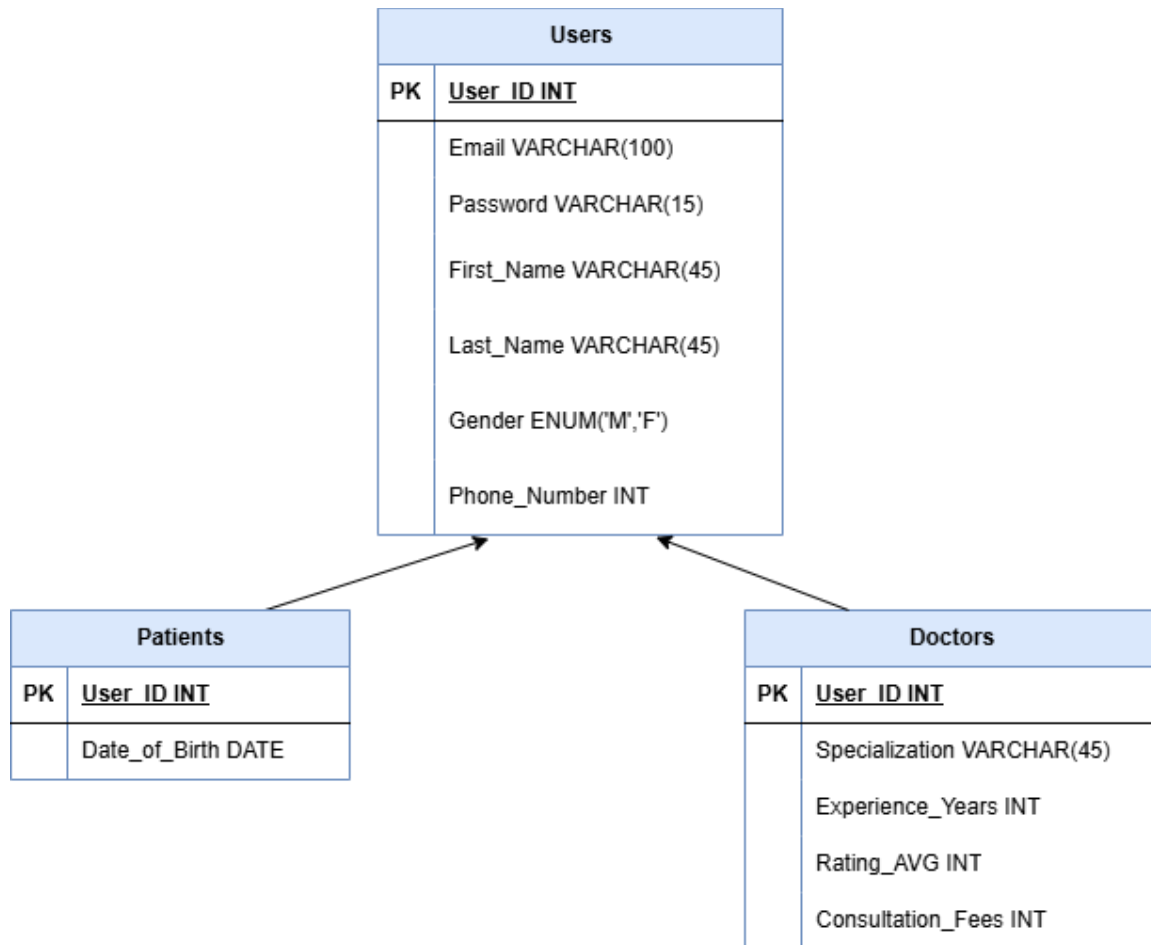
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ER Diagram:-



Explanation of the ISA relationship between Patients & Doctors:



The ER diagram represents a **Smart Medical Appointment Booking System “E7gezly”** that includes several main **Entities**:

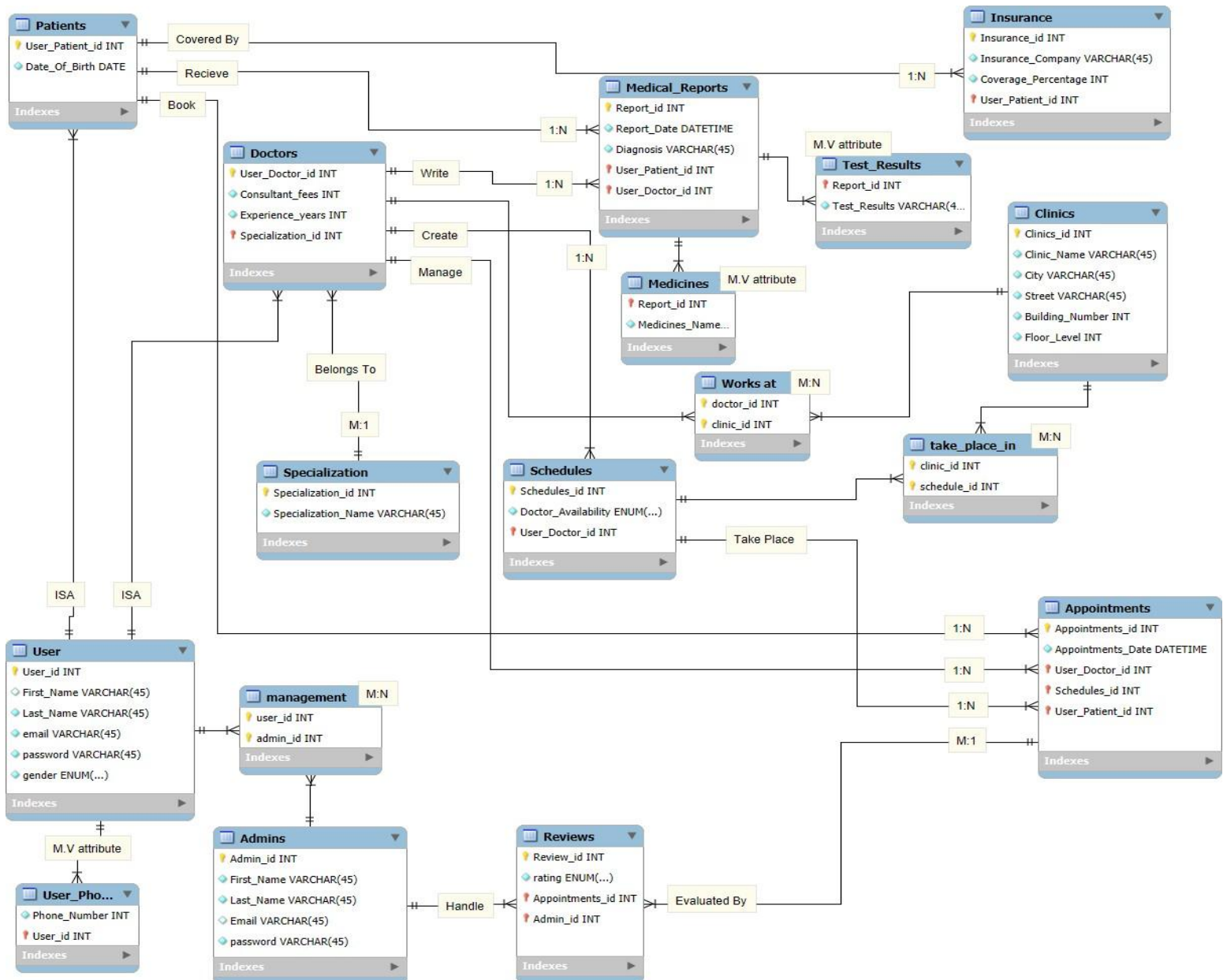
- The User entity is the "parent" entity which has User_ID (PK), Name, Gender, Phone number, Email, Password.
- The Doctor entity inherits the other attributes from "User" stores information such as User_ID (PK, FK), And others like consultation fees, and experience years, rating average and Clinic_ID (FK), Specialization_ID (FK), Schedule_ID (FK), Appointment_ID (FK), Report_ID (FK).
- The specialization entity has Specialization_ID (PK) and Specialization_Name and lastly UserDoctor_ID (FK)
- The Patient entity inherits the other attributes from "User". Patient entity includes User_ID (PK, FK), And others like date of birth and age, Report_ID (FK), Insurance_ID (FK), Appointment_ID (FK).
- The Appointment entity represents the booking between a patient and a doctor and contains Appointment_ID (PK), date, time, status, along with UserPatient_ID (FK) and UserDoctor_ID (FK) to link each appointment to a specific patient and doctor. Reviews_ID (FK), Schedules_ID (FK).
- The Review entity includes Review_ID (PK), rating and uses Admins_ID (FK) and Appointments_ID (FK) to represent that a patient writes a review for a doctor.
- The Medical_Report entity stores medical details such as Report_ID (PK), diagnosis, treatment, date, tests and includes UserPatient_ID (FK) UserDoctor_ID (FK) to associate reports with patients.
- The Schedule entity contains Schedule_ID (PK), available days and times, and includes UserDoctor_ID (FK) to indicate which doctor the schedule belongs to, Clinic_ID (FK) and Appointments_ID (FK).
- The Clinic entity includes Clinic_ID (PK), Name and location also Schedules_ID (FK) and UserDoctors_ID (FK).
- The Insurance entity includes Insurance_ID (PK), company name, and coverage details, and is linked to patients through UserPatient_ID (FK).
- Finally, the Admin entity contains Admin_ID (PK), Name, Password and mail, Also Reviews_ID (FK) and User_ID (FK).

Relationships & 1NF:

The relationships between these entities show that a patient can book multiple appointments, a doctor can handle many appointments, patients can write reviews for doctors only when the appointment is completed, and each doctor operates within a clinic or more and follows a schedule, and there are Admins who handle reviews and manage users, forming a complete and structured healthcare system.

And all the participations of the relationships are “mandatory”.

Regarding normalization, the database satisfies First Normal Form (1NF) because all attributes in each entity are atomic, meaning each field contains only a single value and there are no repeating groups or multi-valued attributes. For example, instead of storing multiple appointments in one field, each appointment is stored as a separate record in the Appointment table, and each attribute (such as date, time, or status) holds only one value. This ensures that the data is going to be organized in a structured tabular form with clear primary keys and no duplicated or nested data within a single column.



2NF & 3NF:

Moving from 1NF to 2NF in this schema eliminates all partial dependencies where non-key attributes were depending on only *part* of a composite primary key rather than the whole key. However, it is worth noting that all tables in the schema already rely on single-column primary keys, which by definition cannot have partial dependencies since there is no composite key to be partially dependent on, meaning the structural difference between 1NF and 2NF here is relatively contained, with the junction tables being the primary area of concern rather than the schema as a whole.

Moving from 2NF to 3NF in this schema eliminates all transitive dependencies, where non-key attributes were depending on other non-key attributes rather than directly on the primary key. Three main changes occur across the schema. First, `rating_avg` is dropped from the `Doctors` table entirely, since it is just a calculated average derivable from the `Reviews` table at any time, meaning storing it directly was creating a risk of stale or inconsistent data. Second, `Coverage_Percentage` is extracted out of the `Insurance` table and moved into a dedicated `Insurance_Company` table, because the coverage percentage is really a property of the company itself rather than each individual insurance record, so if a company changes its rate, one update now covers all patients instead of dozens of rows. Third, `City` is separated from the `Clinics` table into a dedicated `Location` table alongside `Street`, since a city is technically a property of a street rather than an independent fact about the clinic itself, eliminating the risk of the same street being accidentally assigned to two different cities across different records. Together these changes ensure that every attribute on every table describes only and directly its primary key, producing a cleaner, more consistent schema where redundancy is minimized and updates never risk.

Code MySQL:

```
CREATE DATABASE e7gezly;
```

```
USE e7gezly;
```

```
CREATE TABLE Users (
```

```
    User_id INT PRIMARY KEY AUTO_INCREMENT,
```

```
    First_Name VARCHAR(45) NOT NULL,
```

```
    Last_Name VARCHAR(45) NOT NULL,
```

```
    Email VARCHAR(100) NOT NULL UNIQUE,
```

```
Password VARCHAR(255) NOT NULL,  
Gender ENUM('Male','Female') NOT NULL  
);
```

```
CREATE TABLE User_Phone (  
    User_id INT,  
    Phone_Number INT ,  
  
    PRIMARY KEY (User_id, Phone_Number),  
  
    FOREIGN KEY (User_id)  
    REFERENCES Users(User_id)  
);
```

```
CREATE TABLE Patients (  
    User_Patient_id INT PRIMARY KEY,  
    Date_Of_Birth DATE NOT NULL,  
  
    FOREIGN KEY (User_Patient_id)  
    REFERENCES Users(User_id)  
);
```

```
CREATE TABLE Specialization (  
    Specialization_id INT PRIMARY KEY AUTO_INCREMENT,  
    Specialization_Name VARCHAR(45) NOT NULL UNIQUE
```

);

CREATE TABLE Doctors (

User_Doctor_id INT PRIMARY KEY,

Consultant_Fees INT NOT NULL,

Experience_Years INT NOT NULL,

Specialization_id INT NOT NULL,

FOREIGN KEY (User_Doctor_id)

REFERENCES Users(User_id),

FOREIGN KEY (Specialization_id)

REFERENCES Specialization(Specialization_id)

);

CREATE TABLE Admins (

Admin_id INT PRIMARY KEY AUTO_INCREMENT,

First_Name VARCHAR(45) NOT NULL,

Last_Name VARCHAR(45) NOT NULL,

Email VARCHAR(100) NOT NULL UNIQUE,

Password VARCHAR(255) NOT NULL

);

CREATE TABLE Management (

User_id INT,

Admin_id INT,

PRIMARY KEY (User_id, Admin_id),

FOREIGN KEY (User_id)

REFERENCES Users(User_id),

FOREIGN KEY (Admin_id)

REFERENCES Admins(Admin_id)

);

CREATE TABLE Insurance_Company (

Company_id INT PRIMARY KEY AUTO_INCREMENT,

Company_Name VARCHAR(45) NOT NULL,

Coverage_Percentage INT CHECK (Coverage_Percentage BETWEEN 0 AND 100)

);

CREATE TABLE Insurance (

Insurance_id INT PRIMARY KEY AUTO_INCREMENT,

Company_id INT NOT NULL,

User_Patient_id INT NOT NULL,

FOREIGN KEY (Company_id)

REFERENCES Insurance_Company(Company_id),

FOREIGN KEY (User_Patient_id)

REFERENCES Patients(User_Patient_id)

);

```
CREATE TABLE Location (  
    Location_id INT PRIMARY KEY AUTO_INCREMENT,  
    Street VARCHAR(45) NOT NULL,  
    City VARCHAR(45) NOT NULL  
);
```

```
CREATE TABLE Clinics (  
    Clinic_id INT PRIMARY KEY AUTO_INCREMENT,  
    Clinic_Name VARCHAR(45) NOT NULL,  
    Location_id INT NOT NULL,  
    Building_Number INT NOT NULL,  
    Floor_Level INT NOT NULL,  
  
    FOREIGN KEY (Location_id)  
    REFERENCES Location(Location_id)  
);
```

```
CREATE TABLE Works_At (  
    Doctor_id INT,  
    Clinic_id INT,  
  
    PRIMARY KEY (Doctor_id, Clinic_id),  
  
    FOREIGN KEY (Doctor_id)  
    REFERENCES Doctors(User_Doctor_id),
```

```
FOREIGN KEY (Clinic_id)
REFERENCES Clinics(Clinic_id)
);
```

```
CREATE TABLE Schedules (
    Schedule_id INT PRIMARY KEY AUTO_INCREMENT,
    Day_Name VARCHAR(20) NOT NULL,
    Start_Time TIME NOT NULL,
    End_Time TIME NOT NULL,
    User_Doctor_id INT NOT NULL,

    FOREIGN KEY (User_Doctor_id)
    REFERENCES Doctors(User_Doctor_id)
);
```

```
CREATE TABLE Take_Place_In (
    Clinic_id INT,
    Schedule_id INT,

    PRIMARY KEY (Clinic_id, Schedule_id),

    FOREIGN KEY (Clinic_id)
    REFERENCES Clinics(Clinic_id),

    FOREIGN KEY (Schedule_id)
    REFERENCES Schedules(Schedule_id)
);
```

```
CREATE TABLE Appointments (  
    Appointment_id INT PRIMARY KEY AUTO_INCREMENT,  
    Appointment_Date DATETIME NOT NULL,  
    User_Doctor_id INT NOT NULL,  
    Schedule_id INT NOT NULL,  
    User_Patient_id INT NOT NULL,  
  
    FOREIGN KEY (User_Doctor_id)  
    REFERENCES Doctors(User_Doctor_id),  
  
    FOREIGN KEY (Schedule_id)  
    REFERENCES Schedules(Schedule_id),  
  
    FOREIGN KEY (User_Patient_id)  
    REFERENCES Patients(User_Patient_id)  
);
```

```
CREATE TABLE Medical_Reports (  
    Report_id INT PRIMARY KEY AUTO_INCREMENT,  
    Report_Date DATETIME NOT NULL,  
    Diagnosis VARCHAR(100) NOT NULL,  
    User_Patient_id INT NOT NULL,  
    User_Doctor_id INT NOT NULL,  
  
    FOREIGN KEY (User_Patient_id)  
    REFERENCES Patients(User_Patient_id),  
  
    FOREIGN KEY (User_Doctor_id)
```

```
REFERENCES Doctors(User_Doctor_id)
);
```

```
CREATE TABLE Medicines (
    Report_id INT,
    Medicine_Name VARCHAR(45),

    PRIMARY KEY (Report_id, Medicine_Name),

    FOREIGN KEY (Report_id)
    REFERENCES Medical_Reports(Report_id)
);
```

```
CREATE TABLE Test_Results (
    Report_id INT,
    Test_Result VARCHAR(45),

    PRIMARY KEY (Report_id, Test_Result),

    FOREIGN KEY (Report_id)
    REFERENCES Medical_Reports(Report_id)
);
```

```
CREATE TABLE Reviews (
    Review_id INT PRIMARY KEY AUTO_INCREMENT,
    Rating TINYINT CHECK (Rating BETWEEN 1 AND 5),
```

```

Appointment_id INT NOT NULL,
Admin_id INT NOT NULL,

FOREIGN KEY (Appointment_id)
REFERENCES Appointments(Appointment_id),

FOREIGN KEY (Admin_id)
REFERENCES Admins(Admin_id)
);

INSERT INTO Users (First_Name, Last_Name, Email, Password, Gender) VALUES
('Ahmed','Hassan','ahmed.hassan@gmail.com','ah123','Male'),
('Sara','Ali','sara.ali@gmail.com','sa123','Female'),
('Mohamed','Adel','mohamed.adel@gmail.com','mo123','Male'),
('Nour','Khaled','nour.khaled@gmail.com','no123','Female'),
('Omar','Samy','omar.samy@gmail.com','om123','Male'),
('Mariam','Yasser','mariam.yasser@gmail.com','ma123','Female'),
('Youssef','Nabil','youssef.nabil@gmail.com','yo123','Male'),
('Salma','Tarek','salma.tarek@gmail.com','st123','Female'),
('Karim','Fathy','karim.fathy@gmail.com','ka123','Male'),
('Laila','Mostafa','laila.mostafa@gmail.com','la123','Female'),
('Hany','Ibrahim','hany.ibrahim@gmail.com','ha123','Male'),
('Dina','Maher','dina.maher@gmail.com','di123','Female'),
('Amr','Gamal','dr.amr.gamal@e7gezly.com','am123','Male'),
('Reem','Ashraf','dr.reem.ashraf@e7gezly.com','re123','Female'),
('Tamer','Lotfy','dr.tamer.lotfy@e7gezly.com','ta123','Male'),
('Farah','Wael','dr.farah.wael@e7gezly.com','fa123','Female'),
('Khaled','Ragab','dr.khaled.ragab@e7gezly.com','kh123','Male'),
('Menna','Sherif','dr.menna.sherif@e7gezly.com','me123','Female'),

```

```
('Ali','Fouad','dr.ali.fouad@e7gezly.com','al123','Male'),  
('Habiba','Ehab','dr.habiba.ehab@e7gezly.com','hb123','Female');
```

```
INSERT INTO User_Phone VALUES
```

```
(1,'01011111111'),  
(2,'01022222222'),  
(3,'01033333333'),  
(4,'01044444444'),  
(5,'01055555555'),  
(6,'01066666666'),  
(7,'01077777777'),  
(8,'01088888888'),  
(9,'01099999999'),  
(10,'01100000000'),  
(11,'01111111111'),  
(12,'01122222222'),  
(13,'01133333333'),  
(14,'01144444444'),  
(15,'01155555555'),  
(16,'01166666666'),  
(17,'01177777777'),  
(18,'01188888888'),  
(19,'01199999999'),  
(20,'01200000000');
```

```
INSERT INTO Patients VALUES
```

```
(1,'1999-05-10'),
```

(2,'2001-03-15'),
(3,'1998-07-21'),
(4,'2000-11-30'),
(5,'1997-01-25'),
(6,'2002-08-18'),
(7,'1995-12-05'),
(8,'2003-06-09'),
(9,'1996-04-14'),
(10,'2001-09-27'),
(11,'1994-02-11'),
(12,'2000-10-19');

INSERT INTO Specialization (Specialization_Name) VALUES

('Cardiology'),
('Dermatology'),
('Neurology'),
('Pediatrics'),
('Orthopedics');

INSERT INTO Doctors VALUES

(13,500,10,1),
(14,400,7,2),
(15,650,12,3),
(16,300,5,4),
(17,550,9,5),
(18,450,6,1),
(19,350,4,2),

(20,700,15,3);

INSERT INTO Admins (First_Name, Last_Name, Email, Password) VALUES

('Admin','One','admin1@e7gzly.com','admin123'),

('Admin','Two','admin2@e7gzly.com','admin456'),

('Admin','Three','admin3@e7gzly.com','admin789');

INSERT INTO Management VALUES

(1,1),

(2,1),

(13,2),

(14,2),

(15,3);

INSERT INTO Insurance_Company (Company_Name, Coverage_Percentage) VALUES

('AXA',80),

('MetLife',70),

('Allianz',85),

('MedNet',75),

('NextCare',90);

INSERT INTO Insurance (Company_id, User_Patient_id) VALUES

(1,1),

(2,2),

(3,3),

(4,4),

(5,5),

(1,6),
(2,7),
(3,8),
(4,9),
(5,10);

INSERT INTO Location (Street, City) VALUES

('Tahrir Street','Cairo'),
('Nasr Road','Cairo'),
('Corniche','Alexandria'),
('University Street','Mansoura'),
('Sea Road','Hurghada');

INSERT INTO Clinics

(Clinic_Name, Location_id, Building_Number, Floor_Level)

VALUES

('Heart Care Clinic',1,10,2),
('Skin Health Center',2,15,3),
('Brain Clinic',3,20,1),
('Kids Clinic',4,8,2),
('Bone Care Clinic',5,12,4);

INSERT INTO Works_At VALUES

(13,1),
(14,2),
(15,3),
(16,4),
(17,5),

(18,1),
(19,2),
(20,3);

INSERT INTO Schedules

(Day_Name, Start_Time, End_Time, User_Doctor_id)

VALUES

('Sunday','09:00:00','01:00:00',13),
('Monday','10:00:00','02:00:00',14),
('Tuesday','11:00:00','03:00:00',15),
('Wednesday','08:00:00','12:00:00',16),
('Thursday','01:00:00','05:00:00',17),
('Friday','09:00:00','01:00:00',18),
('Saturday','10:00:00','02:00:00',19),
('Sunday','12:00:00','04:00:00',20);

INSERT INTO Take_Place_In VALUES

(1,1),
(2,2),
(3,3),
(4,4),
(5,5),
(1,6),
(2,7),
(3,8);

INSERT INTO Appointments

(Appointment_Date, User_Doctor_id, Schedule_id, User_Patient_id)

VALUES

('2026-05-20 10:00:00',13,1,1),

('2026-05-20 11:00:00',14,2,2),

('2026-05-21 12:00:00',16,4,3),

('2026-05-22 01:00:00',17,5,4),

('2026-05-23 02:00:00',19,7,5),

('2026-05-24 03:00:00',20,8,6);

INSERT INTO Medical_Reports

(Report_Date, Diagnosis, User_Patient_id, User_Doctor_id)

VALUES

('2026-05-20 10:30:00','High Blood Pressure',1,13),

('2026-05-20 11:30:00','Skin Allergy',2,14),

('2026-05-21 12:30:00','Flu',3,16),

('2026-05-22 01:30:00','Knee Pain',4,17),

('2026-05-23 02:30:00','Migraine',5,20);

INSERT INTO Medicines VALUES

(1,'Panadol'),

(1,'Aspirin'),

(2,'Cetirizine'),

(3,'Paracetamol'),

(4,'Ibuprofen'),

(5,'Migranil');

INSERT INTO Test_Results VALUES

(1,'Blood Pressure Test'),

(2,'Skin Test'),

(3,'CBC'),

(4,'X-Ray'),

(5,'MRI');

INSERT INTO Reviews

(Rating, Appointment_id, Admin_id)

VALUES

(5,1,1),

(4,2,2),

(3,3,1),

(5,4,3),

(4,5,2),

(5,6,3);

SELECT

User_Doctor_id,

AVG(Rating) AS Rating_Average

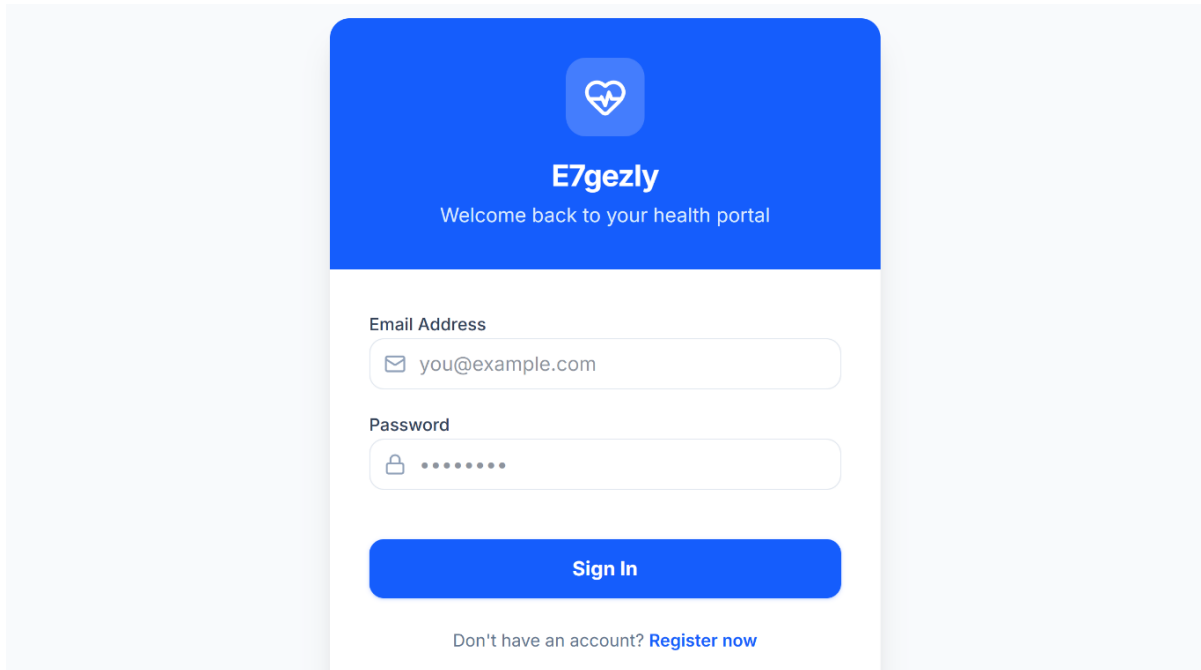
FROM Reviews

JOIN Appointments

ON Reviews.Appointment_id = Appointments.Appointment_id

GROUP BY User_Doctor_id;

➤ Login interface:



The login interface features a blue header with the E7gezly logo and a welcome message. Below the header, there are two input fields for 'Email Address' and 'Password'. The 'Email Address' field contains the text 'you@example.com'. The 'Password' field is masked with dots. A blue 'Sign In' button is positioned below the password field. At the bottom, there is a link to 'Register now' for users who do not have an account.

E7gezly
Welcome back to your health portal

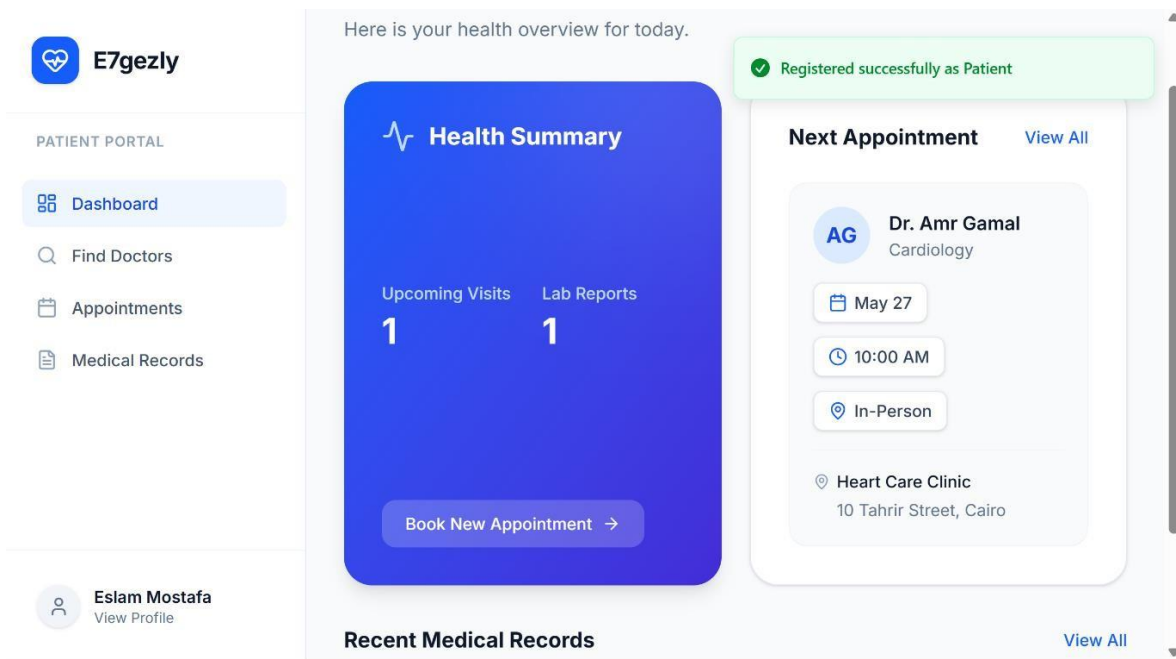
Email Address
you@example.com

Password
••••••••

Sign In

Don't have an account? [Register now](#)

➤ Account Created:



The dashboard is titled 'Here is your health overview for today.' and features a green notification banner stating 'Registered successfully as Patient'. The main content area is divided into two columns. The left column, titled 'Health Summary', shows 'Upcoming Visits' and 'Lab Reports' both with a count of '1' and a 'Book New Appointment' button. The right column, titled 'Next Appointment', shows an appointment with 'Dr. Amr Gamal' in 'Cardiology' on 'May 27' at '10:00 AM' via 'In-Person' at 'Heart Care Clinic'. A 'View All' link is provided for the appointments. The bottom section is titled 'Recent Medical Records' with a 'View All' link. The left sidebar contains a 'PATIENT PORTAL' menu with 'Dashboard', 'Find Doctors', 'Appointments', and 'Medical Records'. The user profile 'Eslam Mostafa' is shown at the bottom left.

E7gezly

PATIENT PORTAL

- Dashboard
- Find Doctors
- Appointments
- Medical Records

Eslam Mostafa
View Profile

Here is your health overview for today.

Registered successfully as Patient

Health Summary

Upcoming Visits: 1
Lab Reports: 1

Next Appointment [View All](#)

AG Dr. Amr Gamal
Cardiology

May 27
10:00 AM
In-Person

Heart Care Clinic
10 Tahrir Street, Cairo

Recent Medical Records [View All](#)

Book New Appointment →

➤ **Booking Appointment:**

➤ **Appointment Booked:**

The screenshot displays the E7gezly Patient Portal interface. On the left, a sidebar contains navigation links: Dashboard, Find Doctors, Appointments (highlighted in blue), and Medical Records. The main content area is titled 'Appointments' and includes a sub-header 'Upcoming'. It features a card for an upcoming appointment with Dr. Amr Gamal in Cardiology on May 27, 2026, at 10:00 AM. The appointment is marked as 'In-Person' and takes place at Heart Care Clinic, 10 Tahrir Street, Cairo. Action buttons for 'Cancel' and 'Get Directions' are visible. Below this, the 'Past Appointments' section shows a completed appointment with Dr. Amr Gamal on 5/20/2026, also in Cardiology and In-Person, marked as 'Completed'.

➤ **Appointment Cancelled:**

 **E7gezly**

PATIENT PORTAL

 Dashboard

 Find Doctors

 **Appointments**

 Medical Records

 **Eslam Mostafa**
View Profile

Appointments

Appointment with Dr. Gamal cancelled

Manage your upcoming visits and view past history.

Upcoming

No upcoming appointments.

Past Appointments

AG

Dr. Amr Gamal

5/27/2026 • Cardiology • In-Person

Cancelled

AG

Dr. Amr Gamal

5/20/2026 • Cardiology • In-Person

Completed